Andrei Stefan Bejgu (he/him)

Date of Birth: 16-11-1995
Location: Rome, Italy
Languages: English (Fluent), Italian (Native), Romanian (Native)

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AI engineer and researcher with **9+ years of experience**. Today I am Senior AI Applied Scientist at Syllotips, working on how its agents plan, use tools, cite sources and improve over time. Before that I spent 8 years at Babelscape, growing from software engineer to NLP researcher, earned an industrial Ph.D. in AI at Sapienza and published at ACL, EMNLP, EACL and NAACL.

Experience & Selected Projects

Syllotips

Remote

Senior AI Applied Scientist Oct. 2025–Present

- Redesigned and rebuilt the **deep-research agent**, the main revenue product: tool calling instead of fixed branches, shared working memory, and a harness with tool budgets and bounded replanning. Judged answer quality rose from 78% to 92% over 192 questions, and from 64% to 95% on customer-style tasks, while follow-up turns got twice as fast.
- Designed the **MCP** layer that connects the agents to enterprise tools such as Atlassian, HubSpot and Intercom: one architecture for cloud and private cloud, OAuth handling, and tool results the agents can cite.
- Built a sandboxed **code interpreter** that lets agents analyse spreadsheets and produce reports and files, on cloud and private cloud alike.
- Rebuilt the **assistant agent** that helps experts close the loop on issues: it investigates what went wrong and proposes a solution for them to review.
- Built **self-hosted model serving** on vLLM across multi-GPU clusters: open-weight LLMs with up to 1M-token context, plus small decision models the agents use for fast routing and grading, so the agents can run fully on our own infrastructure.

Babelscape

Rome, Italy

NLP Researcher Oct. 2021–Sep. 2025

I led advanced NLP research translating state-of-the-art insights into production AI products, with a focus on retrieval-augmented systems and multilingual LLM fine-tuning.

- Designed and developed real-time retrieval pipelines, maximizing precision and efficiency for Vera, a multilingual fact-checking system.
- **Distributed** (Slurm) fine-tuning of open-source LLMs for domain-specific tasks such as **text2sql and claim-extraction/verification**.
- Contributed to the supervised-fine-tuning stage of **Minerva**, the first Italian LLM.

NLP Engineer Jun. 2020–Oct. 2021

- Designed and deployed Semantic Paths, a **Multilingual Semantic Search** engine for scientific articles written in Kotlin.
- Developed advanced **Word Sense Disambiguation** models for large-scale multilingual processing.

Software Engineer Jul. 2017–Jun. 2020

- Developed myKnowledgeGraph, a **Knowledge Graph Extraction** and visualization system for enterprise data.
- Designed the company's **Java NLP Pipeline**, maximizing speed, accuracy, and scalability of multilingual text processing.

CINI - Consorzio Interuniversitario Nazionale per l'Informatica

Rome, Italy

Software Developer Dec. 2016–Jan. 2017

Created an advanced tool for lexicon-semantic stylometric analysis, enabling efficient processing of large-scale texts.

Education

Ph.D. in Artificial Intelligence
Sapienza University of Rome Jan. 2025
Focus: Explored multilingual NLP by applying lexical semantics to modern systems. Improved domain-specific tasks, looked into complex linguistic patterns, and worked on word- and sentence-level representations in RAG systems and LLMs. This was a joint industrial Ph.D., combining academic research with industry-driven projects. Created context-aware, domain-specific embeddings that boosted retrieval performance. –**Advisor:** Prof. R. Navigli

MSc in Computer Science
*Sapienza University of Rome –**Grade**: 110/110 with honors* Nov. 2021
Thesis: Exploring language-agnostic latent spaces for sentences. –**Advisor:** Prof. R. Navigli

BSc in Computer Science
*Sapienza University of Rome –**Grade**: 107/110* Jun. 2019
Thesis: Knowledge-Based Word Sense Disambiguation leveraging synset embeddings. –**Advisor:** Prof. R. Navigli

Publications

[1] A. Scirè*, **A. S. Bejgu***, S. Tedeschi, K. Ghonim, F. Martelli, and R. Navigli. "Truth or Mirage? Towards End-to-End Factuality Evaluation with LLM-OASIS". In: *Computational Linguistics* (2025).

[2] L. Moroni, T. Bonomo, L. Gioffré, L. Xu, D. Fedele, L. Colosi, **A. S. Bejgu**, A. Scirè, and R. Navigli. "What we Learned from Continually Training Minerva: a Case Study on Italian". In: *CLiC-it*. 2025.

[3] L. Moroni, G. Puccetti, P.-L. H. Cabot, **A. S. Bejgu**, A. Miaschi, E. Barba, F. dell'Orletta, A. Esuli, and R. Navigli. "Optimizing LLMs for Italian: Reducing Token Fertility and Enhancing Efficiency Through Vocabulary Adaptation". In: *Findings of NAACL*. 2025.

[4] K. Ghonim, **A. S. Bejgu**, and R. Navigli. "Conceptpedia: Leveraging Traditional Approaches for Multimodal Image-Text Representation". In: *EMNLP*. 2025.

[5] **A. S. Bejgu**, E. Barba, L. Procopio, A. Fernández-Castro, and R. Navigli. "Word Sense Linking: Disambiguating Outside the Sandbox". In: *ACL*. 2024.

[6] F. Molfese, **A. S. Bejgu**, S. Tedeschi, S. Conia, and R. Navigli. "CroCoAlign: A Cross-Lingual, Context-Aware and Fully-Neural Sentence Alignment System for Long Texts". In: *EACL*. 2024.

[7] F. Martelli, **A. S. Bejgu**, et al. "XL-WA: a Gold Evaluation Benchmark for Word Alignment in 14 Language Pairs". In: *CLiC-it*. 2023.

Other Activities

Reviewer Activity: Reviewer for EACL (2023), ACL (2023, external reviewer), EMNLP (2023, external reviewer), ICLR (2024, external reviewer), and ACL ARR (2024 and 2025).
Teaching Activity at Sapienza University of Rome:

- **Natural Language Processing Course (2022–2023)**: Designed and led practical lab sessions and assignments, guiding master's students through state-of-the-art NLP techniques.
- **Hot Topics in NLP (2023–2024)**: Developed and managed one of the three major course assignments, providing hands-on exposure to cutting-edge advancements in NLP.

Scholarships and Awards

- Best Employee Award, Babelscape 2021
- Merit Scholarship Winner, LazioDiSCo.it 2018–2020